

การควบคุมระยะไกลในอุตสาหกรรมและการประมวลผลภาพ

Industrial remote control and image processing

Experiment 5/9: IRIVE Pi Control (2/3) -- M2M Communication

## 5.1 Read More

- <https://th.cytron.io/tutorial/rs485-modbus-rtu-with-iriv-io-controller?srsId=AfmB0orML8whSIUYb6SurlAZFd00l1VLYR9qqK2OfuberT5ymztNuMa7>
- 

## 5.2 เนื้อหาที่เกี่ยวข้อง

### 5.2.1 M2M Communication

### 5.2.2 Modbus RTU with NodeRED

### 5.2.3 Modbus TCP with NodeRED

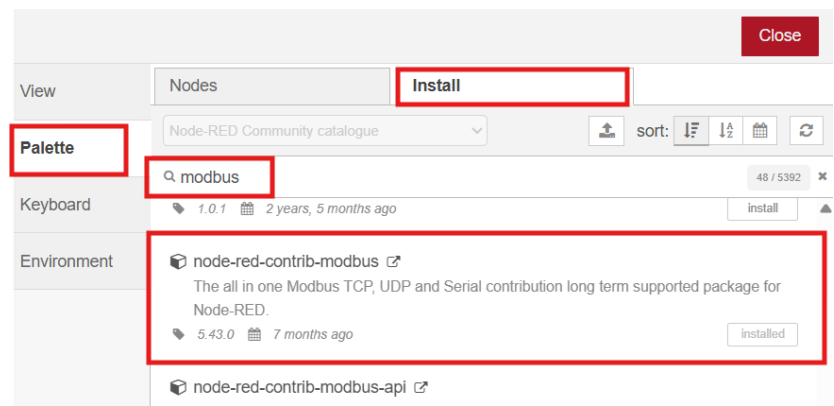
### 5.2.4 RS485 IIoT Gateway

### 5.2.5 Quectel LTE Cat 4 EC25-E Mini PCIe Module

## 5.3 การทดลอง -- M2M Communication

### 5.3.1 Modbus RTU with NodeRED

1. Add Node ➔ Manage Palette

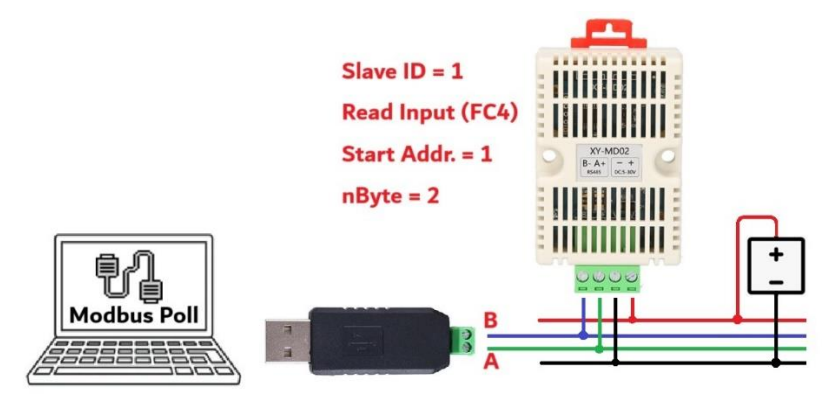


The screenshot shows the Node-RED Manage Palette interface. The 'Nodes' tab is selected, and the 'Install' button is highlighted. The search bar contains 'modbus', and the results list shows 'node-red-contrib-modbus' as the selected node. The node description states it is 'The all in one Modbus TCP, UDP and Serial contribution long term supported package for Node-RED.' and its version is 5.43.0.

- Install
- Palette
- modbus

node-red-contrib-modbus = 5.43.0

2. Test XY-MD02 with Modbus Poll



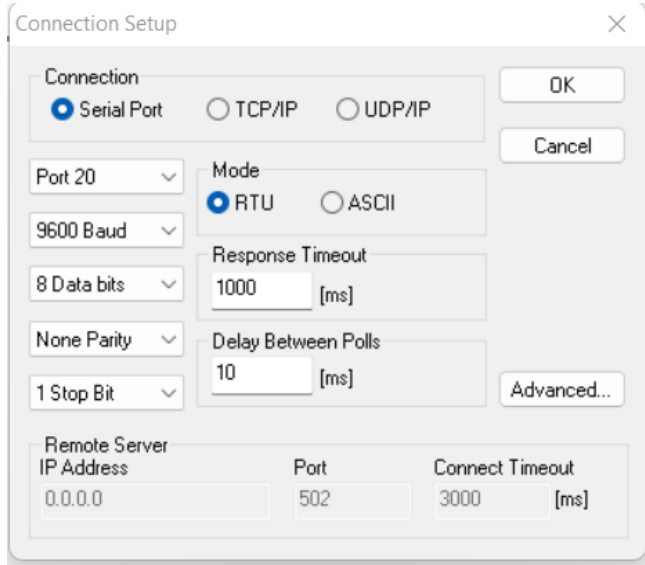
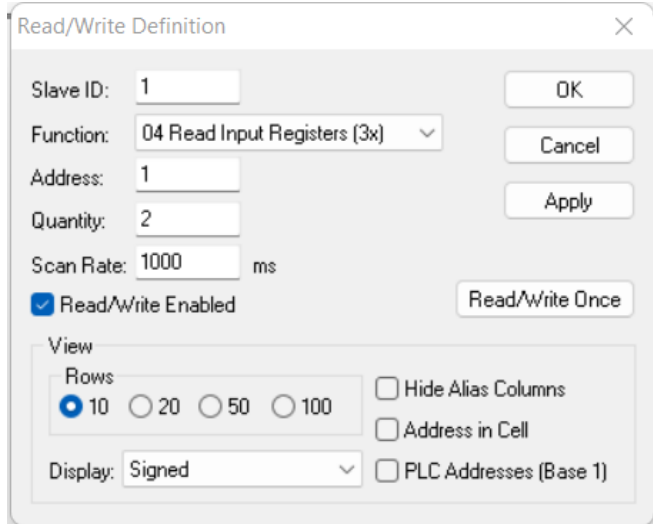
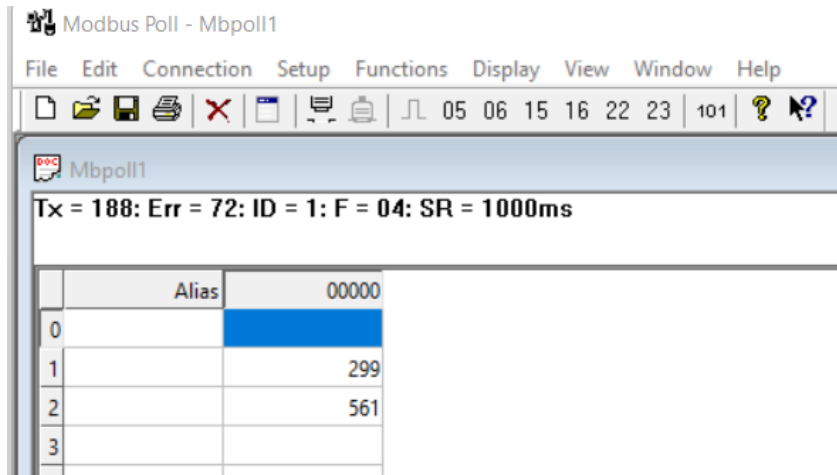
Slave ID = 1  
Read Input (FC4)  
Start Addr. = 1  
nByte = 2

Wiring

โมดูล XY-MD02 มีรีจิสเตอร์ดังนี้

| Type         | Name                   | Size (bytes) | Register Address |
|--------------|------------------------|--------------|------------------|
| Input Reg.   | Temperature            | 2            | 0x0001           |
| Input Reg.   | Humidity               | 2            | 0x0002           |
| Holding Reg. | Device Address         | 2            | 0x0101           |
| Holding Reg. | Baud Rate              | 2            | 0x0102           |
| Holding Reg. | Temperature Correction | 2            | 0x0103           |
| Holding Reg. | Humidity Correction    | 2            | 0x0104           |

Memory Map

|  <p>Connection Setup dialog box showing configuration for a Serial Port connection. The Mode is set to RTU. The Port is 20, Baud rate is 9600, Data bits are 8, Parity is None, and Stop Bit is 1. Response Timeout is 1000 ms and Delay Between Polls is 10 ms. Remote Server IP Address is 0.0.0.0, Port is 502, and Connect Timeout is 3000 ms.</p>                                                                                                             | Read<br>Serial RTU<br>9600,N,8,1                                 |       |       |   |  |       |   |  |     |   |  |     |   |  |  |        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|-------|-------|---|--|-------|---|--|-----|---|--|-----|---|--|--|--------|
|  <p>Read/Write Definition dialog box showing configuration for a Read operation. The Slave ID is 1, Function is 04 Read Input Registers (3x), Address is 1, Quantity is 2, and Scan Rate is 1000 ms. Read/Write Enabled is checked. View settings show 10 rows, Hide Alias Columns is unchecked, Address in Cell is unchecked, and PLC Addresses (Base 1) is unchecked.</p>                                                                                       | ID = 1<br>Function = 04<br>Star Address = 1<br>Quantity = 2 Word |       |       |   |  |       |   |  |     |   |  |     |   |  |  |        |
|  <p>Modbus Poll - Mbpoll1 screenshot showing the main window. The status bar displays: Tx = 188: Err = 72: ID = 1: F = 04: SR = 1000ms. The table below shows the results of the read operation:</p> <table><thead><tr><th></th><th>Alias</th><th>Value</th></tr></thead><tbody><tr><td>0</td><td></td><td>00000</td></tr><tr><td>1</td><td></td><td>299</td></tr><tr><td>2</td><td></td><td>561</td></tr><tr><td>3</td><td></td><td></td></tr></tbody></table> |                                                                  | Alias | Value | 0 |  | 00000 | 1 |  | 299 | 2 |  | 561 | 3 |  |  | Result |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Alias                                                            | Value |       |   |  |       |   |  |     |   |  |     |   |  |  |        |
| 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                  | 00000 |       |   |  |       |   |  |     |   |  |     |   |  |  |        |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                  | 299   |       |   |  |       |   |  |     |   |  |     |   |  |  |        |
| 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                  | 561   |       |   |  |       |   |  |     |   |  |     |   |  |  |        |
| 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                  |       |       |   |  |       |   |  |     |   |  |     |   |  |  |        |

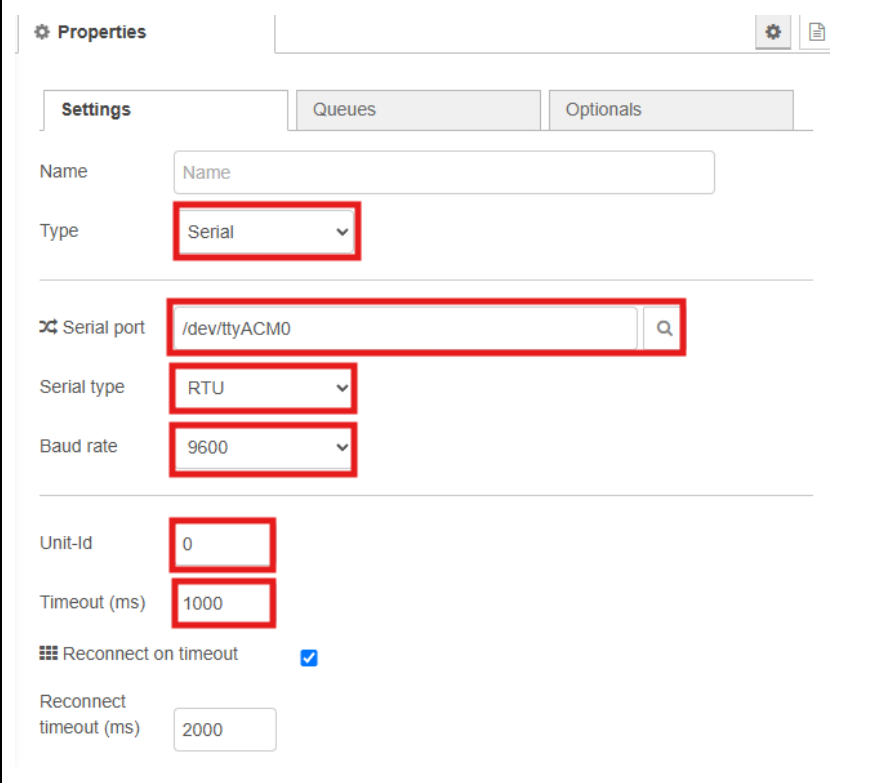
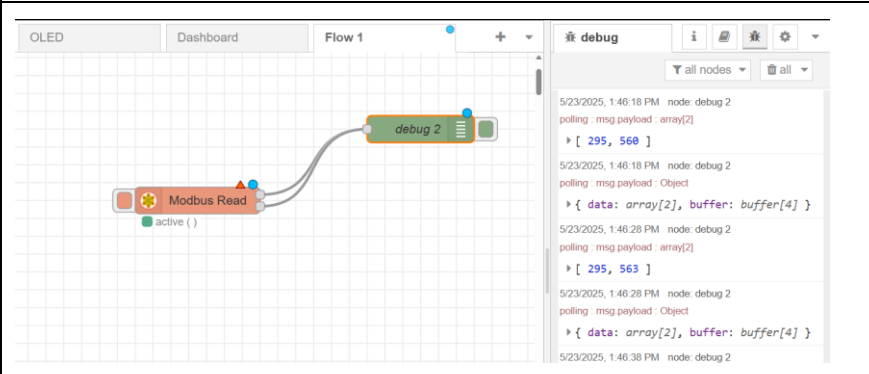
### 3. Test XY-MD02 with Node-RED on IRIV Pi

**Slave ID = 1**  
**Read Input (FC4)**  
**Start Addr. = 1**  
**nByte = 2**

**Wiring**

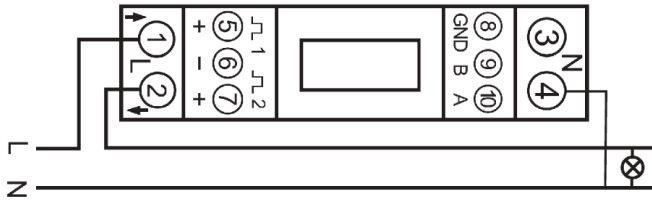
**Place Node**

**Slave ID = 1**  
**Function = 4**  
**Start Addr = 1**  
**Quantity = 2 Byte**  
**Dev/ttyACM0:9600**

|                                                                                     |                                                                                                                       |
|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
|   | <p>Modbus RTU</p> <p>Serial</p> <p>/dev/ttyACM0</p> <p>RTU</p> <p>9600</p> <p>Master ID = 0</p> <p>Timeout = 1000</p> |
|  | <p>Read 2 Data</p>                                                                                                    |

4. Test **SDM-120M** with Node-RED on IRIV Pi

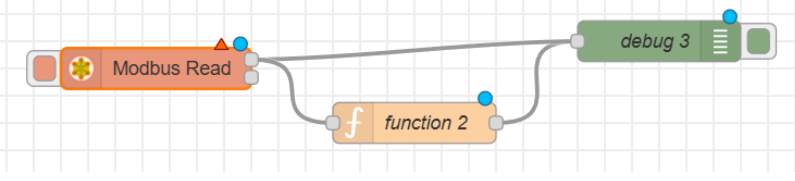
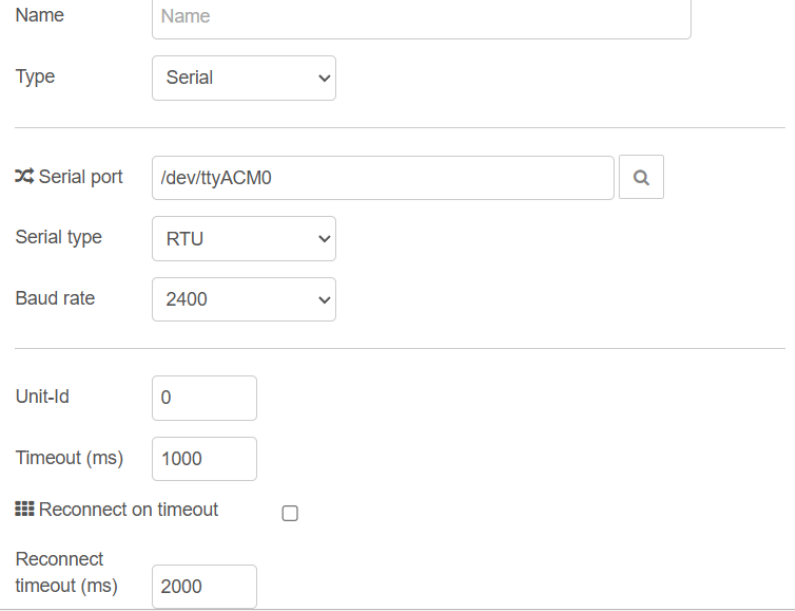
- Baud = 2400(default), 4800, 9600: 8N1

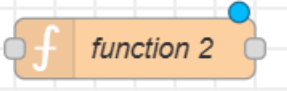
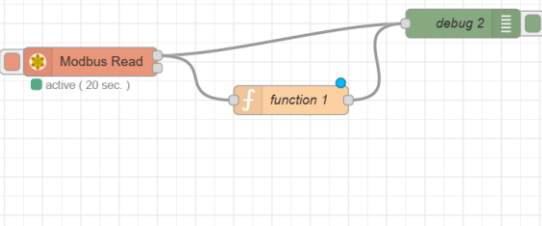


| Address    | Input Register                     | Parameter | Modbus Protocol Start Address Hex |         |         |
|------------|------------------------------------|-----------|-----------------------------------|---------|---------|
| (Register) | Parameters                         | unit      | format                            | Hi byte | Lo byte |
| 30001      | Voltage                            | Volts     | Float                             | 00      | 00      |
| 30007      | Current                            | Amps      | Float                             | 00      | 06      |
| 30013      | Active power                       | Watts     | Float                             | 00      | 0C      |
| 30019      | Apparent power                     | VA        | Float                             | 00      | 12      |
| 30025      | Reactive power                     | VAr       | Float                             | 00      | 18      |
| 30031      | Power factor                       |           | Float                             | 00      | 1E      |
| 30071      | Frequency                          | Hz        | Float                             | 00      | 46      |
| 30073      | Import active energy               | kWh       | Float                             | 00      | 48      |
| 30075      | Export active energy               | kWh       | Float                             | 00      | 4A      |
| 30077      | Import reactive energy             | kvarh     | Float                             | 00      | 4C      |
| 30079      | Export reactive energy             | kvarh     | Float                             | 00      | 4E      |
| 30085      | Total system power demand          | W         | Float                             | 00      | 54      |
| 30087      | Maximum total system power demand  | W         | Float                             | 00      | 56      |
| 30089      | Import system power demand         | W         | Float                             | 00      | 58      |
| 30091      | Maximum Import system power demand | W         | Float                             | 00      | 5A      |
| 30093      | Export system power demand         | W         | Float                             | 00      | 5C      |
| 30095      | Maximum Export system power demand | W         | Float                             | 00      | 5E      |
| 30259      | current demand.                    | Amps      | Float                             | 01      | 02      |
| 30265      | Maximum current demand.            | Amps      | Float                             | 01      | 08      |
| 30343      | Total active energy                | kWh       | Float                             | 01      | 56      |
| 30345      | Total reactive energy              | Kvarh     | Float                             | 01      | 58      |

## Get Voltage

- Read FC=4
- Display Float
- 2 Word
- From 0000H

|                                                                                      |                                                                                                       |
|--------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
|    | Add Node                                                                                              |
|    | ID = 2<br>FC4<br>Address = 0<br>Quantity = 2 Word<br>.<br>Poll Rate = 20 Sec.<br>>> /dev/ttyACM0:2400 |
|  | Type = Serial<br>/dev/ttyACM0<br><br>RTU<br>2400: 8N1<br><br>ID = 0<br>Timeout = 1000                 |

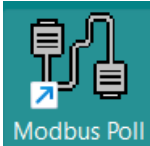
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
|  <pre> 1 // msg.payload contains an array of two 16-bit integers 2 let word1 = msg.payload[1]; // Often the "low word" 3 let word2 = msg.payload[0]; // Often the "high word" 4 5 // Create a 4-byte ArrayBuffer to hold the float 6 let buffer = new ArrayBuffer(4); 7 let view = new DataView(buffer); 8 9 view.setUint16(0, word2, false); // false for Big-Endian (default for Modbus words) 10 view.setUint16(2, word1, false); // false for Big-Endian 11 12 msg.payload = view.getFloat32(0, false); // false for Big-Endian 13 return msg; 14 </pre> <p>// msg.payload contains an array of two 16-bit integers<br/>let word1 = msg.payload[1]; // Often the "low word"<br/>let word2 = msg.payload[0]; // Often the "high word"</p> <p>// Create a 4-byte ArrayBuffer to hold the float<br/>let buffer = new ArrayBuffer(4);<br/>let view = new DataView(buffer);</p> <p>view.setUint16(0, word2, false); // false for Big-Endian (default for Modbus words)<br/>view.setUint16(2, word1, false); // false for Big-Endian</p> <p>msg.payload = view.getFloat32(0, false); // false for Big-Endian<br/>return msg;</p> |        |
|  <pre> [ 17247, 58982 ] 5/26/2025, 4:20:14 PM node: debug 2 polling : msg.payload : number 223.89999389648438 5/26/2025, 4:20:34 PM node: debug 2 polling : msg.payload : array[2] [ 17246, 13107 ] 5/26/2025, 4:20:35 PM node: debug 2 polling : msg.payload : number 222.1999969482422 </pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Deploy |



### 5.3.2 Modbus TCP with NodeRED

#### 5. Test control ModbusTCP

##### 5.1 Modbus TCP Test System

|                                                                                                                                                                                      |                                                                                                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
|  <p>192.168.1.X</p>                                                                                 |                                                                                                       |
|  <p><a href="https://www.modbustools.com/index.html">https://www.modbustools.com/index.html</a></p> |  <p>192.168.1.Y</p> |
| Client                                                                                                                                                                               | Server                                                                                                |

#### 5.2 ติดตั้ง Arduino ESP32 ModbusIP Library

- 5.2.1 from [https://github.com/JoaoCampos89/ModbusIP\\_ESP32/tree/master](https://github.com/JoaoCampos89/ModbusIP_ESP32/tree/master) >> Download [https://github.com/JoaoCampos89/ModbusIP\\_ESP32/archive/refs/heads/master.zip](https://github.com/JoaoCampos89/ModbusIP_ESP32/archive/refs/heads/master.zip)
- 5.2.2 Download <https://github.com/andresarmento/modbus-arduino/archive/refs/heads/master.zip>
- 5.2.3 Sketch → Inc Library → Add Zip ----> Select file “ModbusIP\_ESP32-master.zip”
- 5.2.4 Unzip file “modbus-arduino-master.Zip” copy Modbus libraries to C:\Users\XXXX\Documents\Arduino\libraries

## 5.3 Blink Test ESP32: DOIT ESP32 DevKit Board

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre> 2  #define SW_0 0 3  #define LED_0 2 4 5  void setup() { 6      pinMode(LED_0, OUTPUT); 7  } 8 9  void loop() { 10     digitalWrite(LED_0, HIGH);    delay(100); 11     digitalWrite(LED_0, LOW);     delay(100); 12     digitalWrite(LED_0, HIGH);    delay(100); 13     digitalWrite(LED_0, LOW);     delay(800); 14 } 15 </pre>                                                                                                                                                                                                                                                                                                                                                                                                                 | <pre> #define SW_0 0 #define LED_0 2  void setup() {     pinMode(LED_0, OUTPUT); }  void loop() {     digitalWrite(LED_0, HIGH);    delay(100);     digitalWrite(LED_0, LOW);     delay(100);     digitalWrite(LED_0, HIGH);    delay(100);     digitalWrite(LED_0, LOW);     delay(800); } </pre>                                                                                                                                                                                                                                                                                                                                                                           |
| <pre> 1  #define SW_0 0 2  #define LED_0 2 3  int sts_Mode = 0; 4 5  void setup() { 6      pinMode(SW_0, INPUT_PULLUP); 7      pinMode(LED_0, OUTPUT); 8      Serial.begin(115200); 9  } 10 11 void loop() { 12     if (digitalRead(SW_0) == 0) { 13         sts_Mode = 1 - sts_Mode; 14         delay(50); 15         Serial.println(sts_Mode); 16         while (digitalRead(SW_0) == 0); 17         delay(50); 18     } 19     if (sts_Mode == 1) { 20         digitalWrite(LED_0, HIGH);    delay(100); 21         digitalWrite(LED_0, LOW);     delay(100); 22         digitalWrite(LED_0, HIGH);    delay(100); 23         digitalWrite(LED_0, LOW);     delay(800); 24     } else { 25         digitalWrite(LED_0, HIGH); 26     } 27 } 28 </pre> | <pre> #define SW_0 0 #define LED_0 2  int sts_Mode = 0;  void setup() {     pinMode(SW_0, INPUT_PULLUP);     pinMode(LED_0, OUTPUT);     Serial.begin(115200); }  void loop() {     if (digitalRead(SW_0) == 0) {         sts_Mode = 1 - sts_Mode;         delay(50);         Serial.println(sts_Mode);         while (digitalRead(SW_0) == 0);         delay(50);     }     if (sts_Mode == 1) {         digitalWrite(LED_0, HIGH);    delay(100);         digitalWrite(LED_0, LOW);     delay(100);         digitalWrite(LED_0, HIGH);    delay(100);         digitalWrite(LED_0, LOW);     delay(800);     }     else {         digitalWrite(LED_0, HIGH);     } } </pre> |

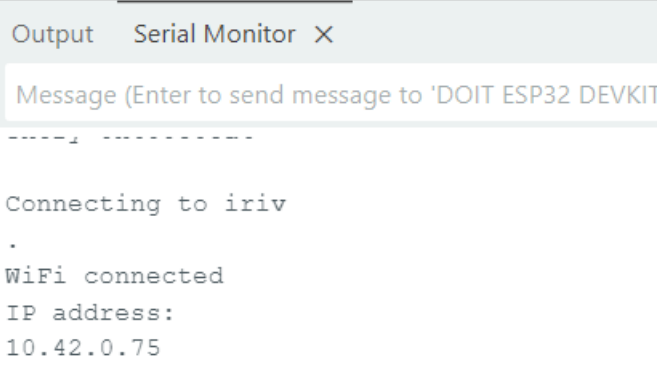
## 5.4 การจองหน่วยความจำ การอ่าน และการเขียนค่า ทำได้ด้วยคำสั่ง ดังนี้

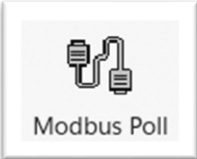
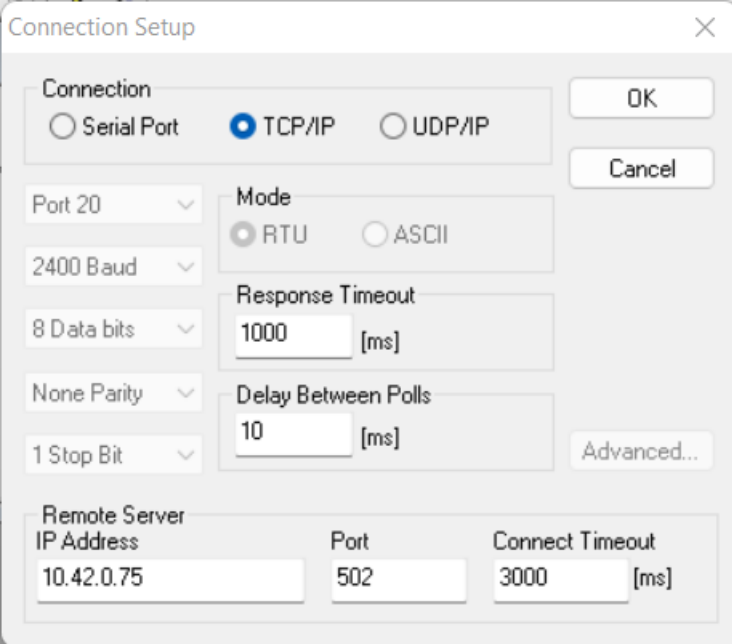
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre> 1  // allocate memory in setup 2  mb.addHreg(Addr0, Data); // allocate Holding register + Store Data 3  mb.addIreg(Addr1);       // allocate Inputs register 4  mb.addCoil(Addr2);       // allocate Coil input/output 5  md.addIsts(Addr3);       // allocate Digital inputs 6 7  // Write data to memory for Client Read 8  mb.Hreg(Addr0, Value);   // Write Holding register 9  mb.Iregs(Addr1, Value);  // Write Inputs register 10 mb.Coils(Addr2, 1);      // Write Coil input/output 11 mb.Ists(Addr3, 1);       // Write Digital inputs 12 13 // From Client Write -&gt; Read data from memory 14 int16_t holdingRegValue = mb.Hreg(Addr0); // Read Holding register 15 byte coilsValue = mb.Coil(Addr2);         // Read Coil input/output </pre> | <pre> // allocate memory in setup mb.addHreg(Addr0, Data); // allocate Holding register + Store Data mb.addIreg(Addr1);       // allocate Inputs register mb.addCoil(Addr2);       // allocate Coil input/output md.addIsts(Addr3);       // allocate Digital inputs  // Write data to memory for Client Read mb.Hreg(Addr0, Value);   // Write Holding register mb.Iregs(Addr1, Value);  // Write Inputs register mb.Coils(Addr2, 1);      // Write Coil input/output mb.Ists(Addr3, 1);       // Write Digital inputs  // From Client Write -&gt; Read data from memory int16_t holdingRegValue = mb.Hreg(Addr0); // Read Holding register byte coilsValue = mb.Coil(Addr2);         // Read Coil input/output </pre> |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## 5.5 Read/Write Coils

PC: Modbus Poll  $\longleftrightarrow$  ESP32(Server)

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre> 1  /* 2  Modbus-Arduino Example - Test Led (Modbus) 3  Control a Led on GPIO0 pin using Write Single Coil Modbus 4  Copyright by André Sarmento Barbosa 5  http://github.com/andresarmiento/modbus-arduino 6  */ 7 8  #include &lt;Modbus.h&gt; 9  #include &lt;ModbusIP_ESP32.h&gt; 10 11 #define mySSID "iriv" 12 #define myPass "iriv1234" 13 14 //Modbus Registers Offsets (0-9999) 15 const int LED_COIL = 100; 16 //Used Pins 17 const int ledPin = 2; // LED Test 18 19 //ModbusIP object 20 ModbusIP mb; 21 22 void setup() { 23   Serial.begin(115200); 24   Serial.print("\nConnecting to "); 25   Serial.println(mySSID); 26   mb.config(mySSID, myPass); 27   while (WiFi.status() != WL_CONNECTED) { 28     delay(500); 29     Serial.print("."); 30   } 31   Serial.println(""); 32   Serial.println("WiFi connected"); 33   Serial.println("IP address: "); 34   Serial.println(WiFi.localIP()); 35 36   pinMode(ledPin, OUTPUT); 37   mb.addCoil(LED_COIL); 38 } 39 40 void loop() { 41   //Call once inside loop() - all magic here 42   mb.task(); 43   //Attach ledPin to LED_COIL register 44   digitalWrite(ledPin, mb.Coil(LED_COIL)); 45 } 46 </pre> | <pre> /* Modbus-Arduino Example - Test Led (Modbus IP ESP32) Control a Led on GPIO0 pin using Write Single Coil Modbus Function Copyright by André Sarmento Barbosa http://github.com/andresarmiento/modbus-arduino */  #include &lt;Modbus.h&gt; #include &lt;ModbusIP_ESP32.h&gt;  #define mySSID "IoT-Test" #define myPass "212224236248"  //Modbus Registers Offsets (0-9999) const int LED_COIL = 100; //Used Pins const int ledPin = 2; // LED Test  //ModbusIP object ModbusIP mb;  void setup() {   Serial.begin(115200);   Serial.print("\nConnecting to ");   Serial.println(mySSID);   mb.config(mySSID, myPass);   while (WiFi.status() != WL_CONNECTED) {     delay(500);     Serial.print(".");   }    Serial.println("");   Serial.println("WiFi connected");   Serial.println("IP address: ");   Serial.println(WiFi.localIP());    pinMode(ledPin, OUTPUT);   mb.addCoil(LED_COIL); }  void loop() {   //Call once inside loop() - all magic here   mb.task();    //Attach ledPin to LED_COIL register   digitalWrite(ledPin, mb.Coil(LED_COIL)); } </pre> |
| <div>Output Serial Monitor X</div> <div>Not connected. Select a board and a port to connect automatically.</div> <div>Connecting to iriv</div> <div>.</div> <div>WiFi connected</div> <div>IP address:</div> <div>10.42.0.75</div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

|                                                                                                                                                                                                                                                                                           |                                                                                      |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
|  <p>Output Serial Monitor X</p> <p>Message (Enter to send message to 'DOIT ESP32 DEVKIT')</p> <p>-----</p> <p>Connecting to iriv</p> <p>.</p> <p>WiFi connected</p> <p>IP address:</p> <p>10.42.0.75</p> | <p>Arduino IDE: Serial Monitor</p> <p>&gt;&gt; Connect AP</p> <p>&gt;&gt; Get IP</p> |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p>Modbus Poll</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <p>Start Modbus Poll</p>                                                                                                                                       |
|  <p>Connection Setup</p> <p>Connection</p> <p><input type="radio"/> Serial Port <input checked="" type="radio"/> TCP/IP <input type="radio"/> UDP/IP</p> <p>Port 20 Mode</p> <p>2400 Baud <input checked="" type="radio"/> RTU <input type="radio"/> ASCII</p> <p>8 Data bits Response Timeout</p> <p>None Parity 1000 [ms]</p> <p>1 Stop Bit Delay Between Polls</p> <p>10 [ms]</p> <p>Advanced...</p> <p>Remote Server</p> <p>IP Address Port Connect Timeout</p> <p>10.42.0.75 502 3000 [ms]</p> | <p>Modbus Poll – Connection Setup</p> <p>Modbus TCP/IP</p> <p>IP Server = xx.xx.xx.xx</p> <p>Server Port = 502</p> <p>Timeout = 3000</p> <p>IP Mode = IPv4</p> |

Read/Write Definition

Slave ID:  OK Cancel

Function:

Address mode  
☒ Dec ☐ Hex

Address:  PLC address = 00101

Quantity:

Scan Rate:  [ms] Apply

Disable  
☐ Read/Write Disabled  
☐ Disable on error Read/Write Once

View  
 Rows  
☒ 10 ☐ 20 ☐ 50 ☐ 100 ☐ Fit to Quantity  
☐ Hide Name Columns ☐ PLC Addresses (Base 1)  
☐ Address in Cell ☐ Enron/Daniel Mode

Request  
 RTU 01 01 00 64 00 0A FD D2  
 ASCII 3A 30 31 30 31 30 30 36 34 30 30 30 41 39 30 0D 0A

## Read/Write Definition

Slave ID = 1  
 Func = 01 Read Coils  
 Address Mode = Dec  
 Address = 100  
 Quantity = 10  
 Scan Rate = 1000  
 View Row = 10

Modbus Poll - Mbpoll1

File Edit Connection Setup Functions

05 06 15

Mbpoll1

Tx = 72: Err = 46: ID = 1: F = 01: SR =

|   | Name | 00100 |
|---|------|-------|
| 0 |      | 1     |
| 1 |      | 0     |
| 2 |      | 0     |
| 3 |      | 0     |
| 4 |      | 0     |
| 5 |      | 0     |
| 6 |      | 0     |
| 7 |      | 0     |

Write Single Coil

Slave ID:  Send Cancel

Address:

Value  
☒ On ☐ Off

Result  
 Response ok  
☐ Close dialog on "Response ok"

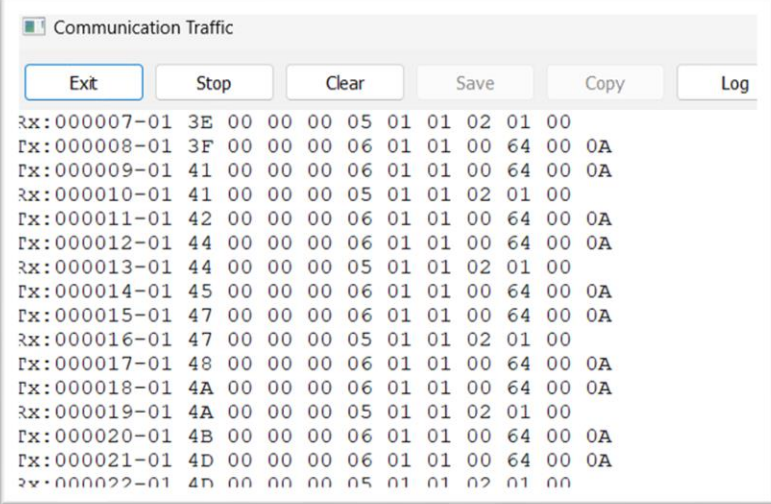
Use Function  
☒ 05: Write single coil  
☐ 15: Write multiple coils

Request  
 RTU 01 05 00 64 FF 00 CD E5

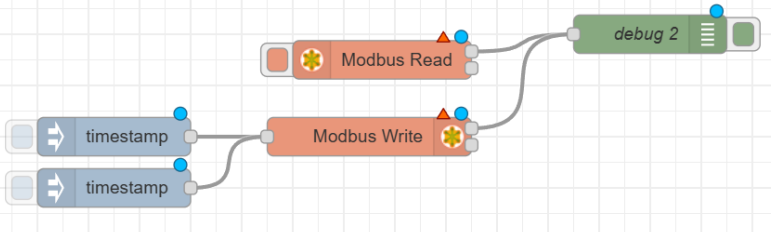
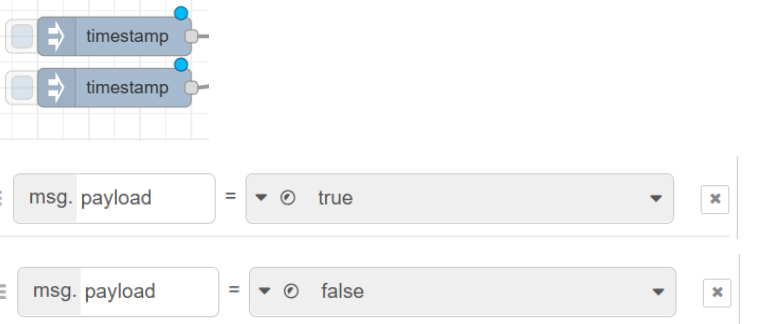
## Double Click 100

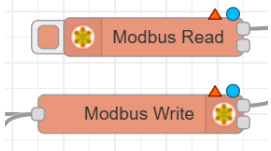
Slave ID = 1  
 Address = 100  
 Value On  
 Off

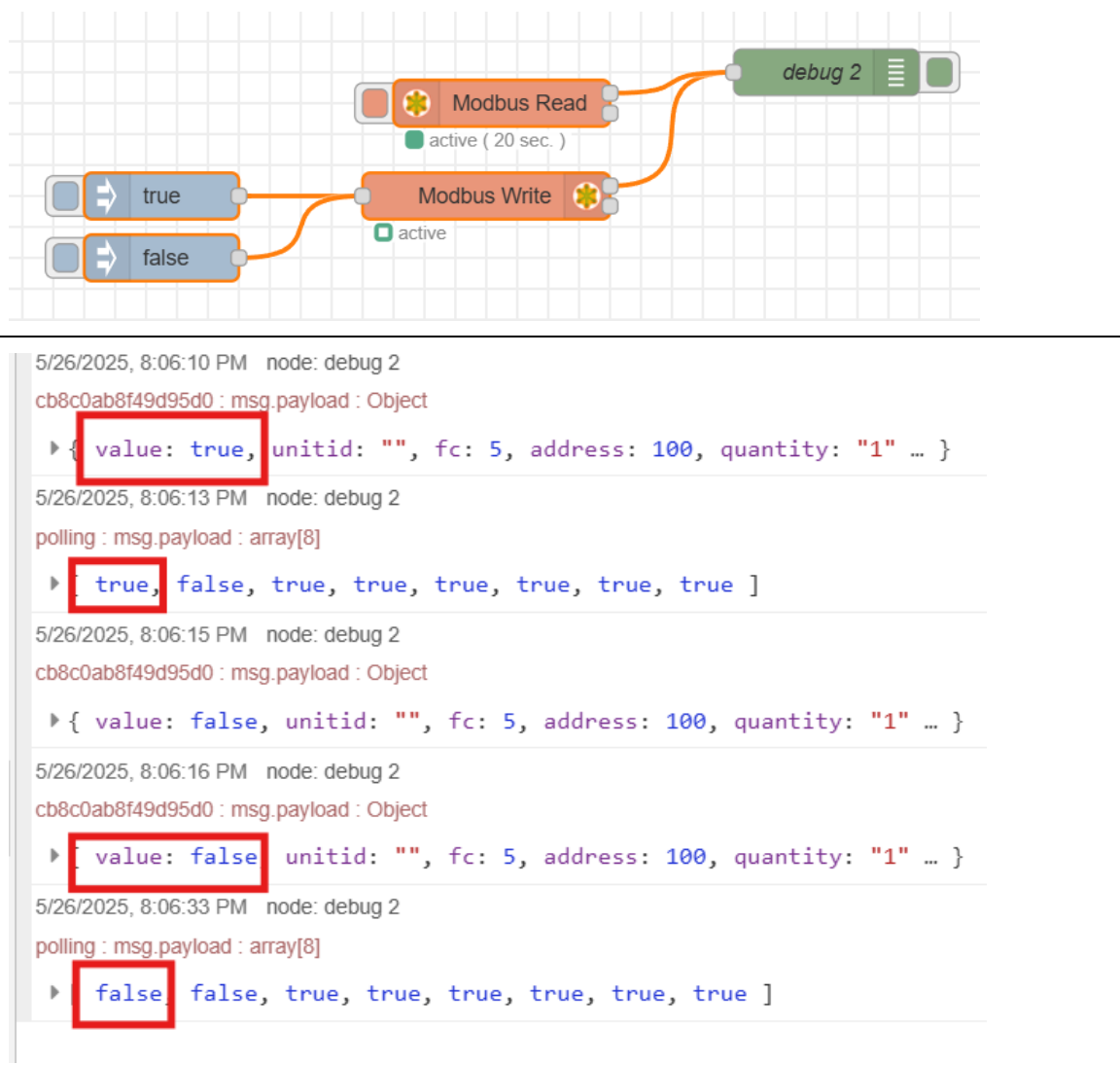
Send

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
|  <pre> Rx:000007-01 3E 00 00 00 05 01 01 02 01 00 Tx:000008-01 3F 00 00 00 06 01 01 00 64 00 0A Tx:000009-01 41 00 00 00 06 01 01 00 64 00 0A Rx:000010-01 41 00 00 00 05 01 01 02 01 00 Tx:000011-01 42 00 00 00 06 01 01 00 64 00 0A Tx:000012-01 44 00 00 00 06 01 01 00 64 00 0A Rx:000013-01 44 00 00 00 05 01 01 02 01 00 Tx:000014-01 45 00 00 00 06 01 01 00 64 00 0A Tx:000015-01 47 00 00 00 06 01 01 00 64 00 0A Rx:000016-01 47 00 00 00 05 01 01 02 01 00 Tx:000017-01 48 00 00 00 06 01 01 00 64 00 0A Tx:000018-01 4A 00 00 00 06 01 01 00 64 00 0A Rx:000019-01 4A 00 00 00 05 01 01 02 01 00 Tx:000020-01 4B 00 00 00 06 01 01 00 64 00 0A Tx:000021-01 4D 00 00 00 06 01 01 00 64 00 0A Rx:000022-01 4D 00 00 00 05 01 01 02 01 00 </pre> | Communication Traffic |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|

### 5.6 Read/Write Coil with Node-Red from IRIV Pi

|                                                                                      |                                                 |
|--------------------------------------------------------------------------------------|-------------------------------------------------|
|   |                                                 |
|  | Inject<br><br>Boolean = True<br>Boolean = False |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                          |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p><b>Settings</b> Queues Optionals</p> <p>Name <input type="text" value="Name"/></p> <p>Type <input type="text" value="TCP"/></p> <hr/> <p>Host <input type="text" value="10.42.0.75"/></p> <p>Port <input type="text" value="502"/></p> <p>TCP Type <input type="text" value="DEFAULT"/></p> <hr/> <p>Unit-Id <input type="text" value="0"/></p> <p>Timeout (ms) <input type="text" value="1000"/></p> <p><input checked="" type="checkbox"/> Reconnect on timeout</p> | <p>Modbus Read<br/>Modbus Write</p> <p>Set Server</p> <p>Type = TCP<br/>Host Server IP = xx.xx.xx.xx<br/>Port = 502</p>                                  |
| <p><b>Settings</b> Optionals</p> <p>Name <input type="text" value="Name"/></p> <p>Topic <input type="text" value="Topic"/></p> <p>Unit-Id <input type="text"/></p> <p>FC <input type="text" value="FC 1: Read Coil Status"/></p> <p>Address <input type="text" value="100"/></p> <p>Quantity <input type="text" value="1"/></p> <p>Poll Rate <input type="text" value="1"/> <input type="text" value="second(s)"/></p> <p><input type="checkbox"/> Delay to activate input</p> <p>Server <input type="text" value="modbus-tcp@10.42.0.75:502"/></p>        | <p>Read Modbus</p> <p>FC1 – Read Single Coil<br/>Address = 100<br/>Quantity = 1</p> <p>Poll Rate = 1 Sec.</p> <p>Server = modbus-tcp@xx.xx.xx.xx:502</p> |
| <p><b>Settings</b> Optionals</p> <p>Name <input type="text" value="Name"/></p> <p>Unit-Id <input type="text"/></p> <p>FC <input type="text" value="FC 5: Force Single Coil"/></p> <p>Address <input type="text" value="100"/></p> <p><input type="checkbox"/> Delay to activate input</p> <p>Server <input type="text" value="modbus-tcp@10.42.0.75:502"/></p>                                                                                                                                                                                             | <p>Write Modbus</p> <p>FC5 – Write Single Coil<br/>Address = 100</p> <p>Server = modbus-tcp@xx.xx.xx.xx:502</p>                                          |



The image shows a Node-RED flow diagram and its corresponding debug console output. The flow diagram at the top consists of two input nodes labeled 'true' and 'false', each with a switch icon. These are connected to a 'Modbus Write' node, which is marked as 'active'. The 'Modbus Write' node is then connected to a 'Modbus Read' node, also marked as 'active (20 sec.)'. Finally, the 'Modbus Read' node is connected to a 'debug 2' node.

The debug console below shows the following sequence of messages:

```
5/26/2025, 8:06:10 PM node: debug 2
cb8c0ab8f49d95d0 : msg.payload : Object
  ▶ { value: true, unitid: "", fc: 5, address: 100, quantity: "1" ... }

5/26/2025, 8:06:13 PM node: debug 2
polling : msg.payload : array[8]
  ▶ [ true, false, true, true, true, true, true, true ]

5/26/2025, 8:06:15 PM node: debug 2
cb8c0ab8f49d95d0 : msg.payload : Object
  ▶ { value: false, unitid: "", fc: 5, address: 100, quantity: "1" ... }

5/26/2025, 8:06:16 PM node: debug 2
cb8c0ab8f49d95d0 : msg.payload : Object
  ▶ [ value: false unitid: "", fc: 5, address: 100, quantity: "1" ... }

5/26/2025, 8:06:33 PM node: debug 2
polling : msg.payload : array[8]
  ▶ [ false, false, true, true, true, true, true, true ]
```

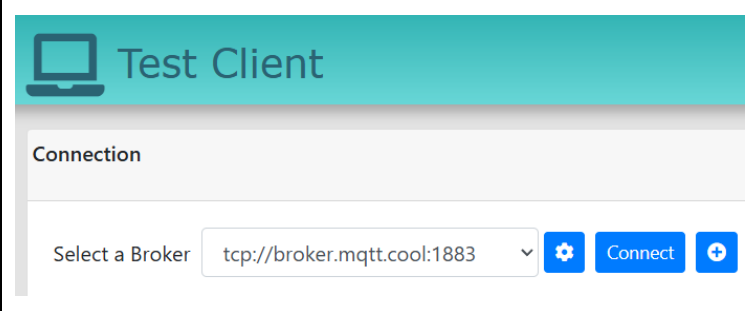
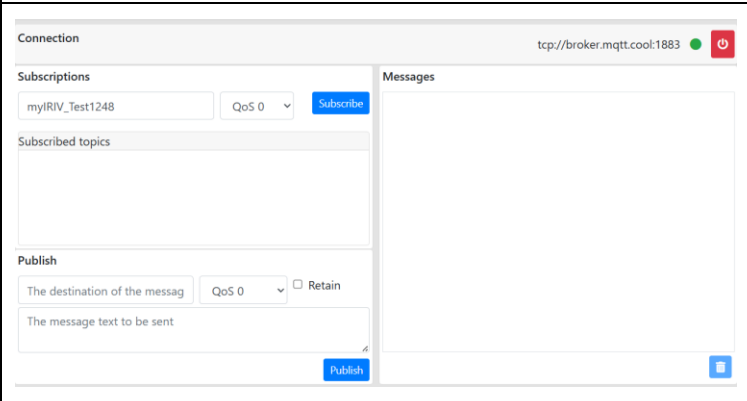
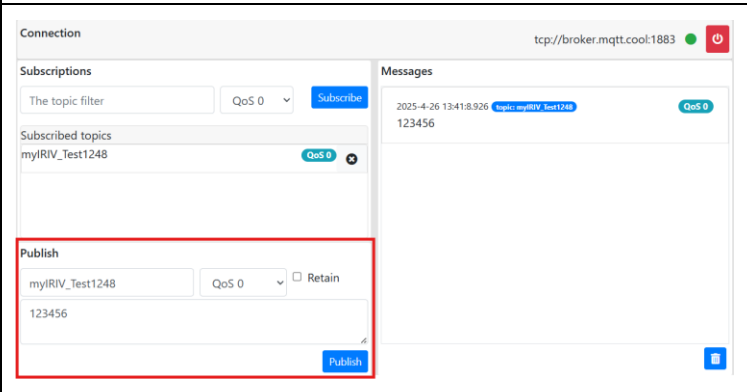
In the debug console, the 'value' property of the first and third JSON objects, and the first element of the second and fourth arrays, are highlighted with red boxes, showing a sequence of true, false, true, false.



### 5.3.3 RS485 IIoT Gateway

#### 6. Test Read SDM120M to MQTT Broker / Control from Broker

##### 6.1 Set Broker >> <https://testclient-cloud.mqtt.cool/>

|                                                                                     |                                                                                                                          |
|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
|    | <a href="https://testclient-cloud.mqtt.cool/">https://testclient-cloud.mqtt.cool/</a><br><br>tcp://broker.mqtt.cool:1883 |
|   | Subscribe topic:<br>myIRIV_Test1248                                                                                      |
|  | Publish to topic<br>myIRIV_Test1248                                                                                      |

## 6.2 Read SDM120M and Send to Broker

|                           |                                                                                      |
|---------------------------|--------------------------------------------------------------------------------------|
|                           |                                                                                      |
| <p>active ( 30 sec. )</p> | <p>Modbus Read &lt; last topic &gt;<br/>Function &lt; last topic &gt;</p>            |
|                           | <p>Add Server</p> <p>Name = myTest<br/>Server = broker.mqtt.cool<br/>Port = 1883</p> |
|                           | <p>Server = myTest<br/>Topic = myIRIV_Test1248</p>                                   |

```

graph LR
    ModbusRead[Modbus Read] --> function1[function 1]
    function1 --> debug2[debug 2]
    function1 --> myIRIV_Test1248[myIRIV_Test1248]
    ModbusRead -- active (30 sec.)
    debug2 -- connected
  
```

5/26/2025, 8:43:40 PM node: debug 2  
polling : msg.payload : number  
219.60000610351562

5/26/2025, 8:43:57 PM node: debug 2  
polling : msg.payload : number  
220.39999389648438

tcp://broker.mqtt.cool:1883

**Messages**

|                        |                        |       |
|------------------------|------------------------|-------|
| 2025-4-26 13:44:27.366 | topic: myIRIV_Test1248 | QoS 0 |
| 220                    |                        |       |
| 2025-4-26 13:43:57.370 | topic: myIRIV_Test1248 | QoS 0 |
| 220.39999389648438     |                        |       |
| 2025-4-26 13:43:41.76  | topic: myIRIV_Test1248 | QoS 0 |
| 219.60000610351562     |                        |       |
| 2025-4-26 13:41:8.926  | topic: myIRIV_Test1248 | QoS 0 |
| 123456                 |                        |       |

Data Show on Broker

### 5.3.4 Quectel LTE Cat 4 EC25-E Mini PCIe Module

#### 7. Start EC25 on IRIV PiControl

##### 7.1 ติดตั้งบอร์ดเข้ากับ IRIV Pi

##### 7.2 เปิด IRIV Pi ตรวจสอบว่าตัวบอร์ด Raspberry Pi มองเห็น 4G module ที่เราได้ติดตั้งเข้าไปหรือไม่ และ 4G module ทำงานอยู่ในโหมดใดด้วยคำสั่ง

**lsusb -t**

```
cytron@pi5:~$ lsusb -t
/: Bus 04.Port 1: Dev 1, Class=root_hub, Driver=xhci-hcd/1p, 5000M
/: Bus 03.Port 1: Dev 1, Class=root_hub, Driver=xhci-hcd/2p, 480M
/: Bus 02.Port 1: Dev 1, Class=root_hub, Driver=xhci-hcd/1p, 5000M
/: Bus 01.Port 1: Dev 1, Class=root_hub, Driver=xhci-hcd/2p, 480M
   |__ Port 1: Dev 6, If 0, Class=Vendor Specific Class, Driver=option, 480M
   |__ Port 1: Dev 6, If 1, Class=Vendor Specific Class, Driver=option, 480M
   |__ Port 1: Dev 6, If 2, Class=Vendor Specific Class, Driver=option, 480M
   |__ Port 1: Dev 6, If 3, Class=Vendor Specific Class, Driver=option, 480M
   |__ Port 1: Dev 6, If 4, Class=Vendor Specific Class, Driver=qmi_wwan, 480M
   |__ Port 2: Dev 2, If 0, Class=Human Interface Device, Driver=usbhid, 12M
   |__ Port 2: Dev 2, If 1, Class=Human Interface Device, Driver=usbhid, 12M
cytron@pi5:~$
```

โดยหากทำงานอยู่ในโหมด QMI จะปรากฏข้อความดังภาพ

```
cytron@pi5:~$ lsusb -t
/: Bus 04.Port 1: Dev 1, Class=root_hub, Driver=xhci-hcd/1p, 5000M
/: Bus 03.Port 1: Dev 1, Class=root_hub, Driver=xhci-hcd/2p, 480M
/: Bus 02.Port 1: Dev 1, Class=root_hub, Driver=xhci-hcd/1p, 5000M
/: Bus 01.Port 1: Dev 1, Class=root_hub, Driver=xhci-hcd/2p, 480M
   |__ Port 1: Dev 5, If 0, Class=Vendor Specific Class, Driver=option, 480M
   |__ Port 1: Dev 5, If 1, Class=Vendor Specific Class, Driver=option, 480M
   |__ Port 1: Dev 5, If 2, Class=Vendor Specific Class, Driver=option, 480M
   |__ Port 1: Dev 5, If 3, Class=Vendor Specific Class, Driver=option, 480M
   |__ Port 1: Dev 5, If 4, Class=Communications, Driver=cdc_ether, 480M
   |__ Port 1: Dev 5, If 5, Class=CDC Data, Driver=cdc_ether, 480M
   |__ Port 2: Dev 2, If 0, Class=Human Interface Device, Driver=usbhid, 12M
   |__ Port 2: Dev 2, If 1, Class=Human Interface Device, Driver=usbhid, 12M
cytron@pi5:~$
```

โดยหากทำงานอยู่ในโหมด ECM จะปรากฏข้อความดังภาพ

โดยหาก 4G module ทำงานใน ECM mode สามารถข้ามไปขั้นตอนที่ 7.4 ได้เลย

##### 7.3 (ทำเฉพาะกรณีที 4G module ทำงานในโหมด QMI): ทำการดาวน์โหลดเครื่องมือสำหรับเปลี่ยนโหมดการทำงานของ 4G module ด้วยคำสั่งด้านล่าง (พิมพ์ที่ละบรรทัด และ ระหว่างทำขั้นตอนนี้บอร์ดจะต้องมีอินเทอร์เน็ต เพราะต้องดาวน์โหลดเครื่องมือจากอินเทอร์เน็ต)

**sudo apt-get install minicom**

หลังจากนั้นใช้คำสั่ง

**sudo minicom -D /dev/ttyUSB2**

โดยจะพบกับหน้าต่างลักษณะดังภาพ ให้พิมพ์คำว่า AT (ตัวพิมพ์ใหญ่) แล้วกด Enter ซึ่งถ้าหากโมดูลตอบสนอง จะปรากฏคำว่า OK ขึ้นมา

หมายเหตุ ในการพิมพ์คำสั่ง ถือเป็นเรื่องปกติที่จะไม่มีตัวอักษรปรากฏขึ้นมาขณะพิมพ์ โดยสามารถแก้ไขได้โดยการพิมพ์คำสั่ง (ตัวพิมพ์ใหญ่)

|                           |                                                                                                                                                                                          |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ATE1</b>               | แล้วกด Enter เพียงเท่านี้ คำสั่งต่อไปก็จะปรากฏขึ้นมาให้เห็นตอนเราพิมพ์แล้ว<br>*กรณี 4G module มีการ restart หรือเริ่มการทำงานใหม่ ผลของคำสั่ง ATE1 จะหายไป และต้องเปิดขึ้นมาใหม่อีกครั้ง |
|                           | หลังจากนั้น ใช้คำสั่ง (ที่ละบรรทัด แล้วกด Enter)                                                                                                                                         |
| <b>AT+QCFG="usbnet",1</b> |                                                                                                                                                                                          |
| <b>AT+CFUN=1,1</b>        | โดยหลังจากนั้น ตัวโมดูลจำการรีสตาร์ทตัวเองโดยอัตโนมัติ และจะทำงานในโหมด ECM แทน                                                                                                          |

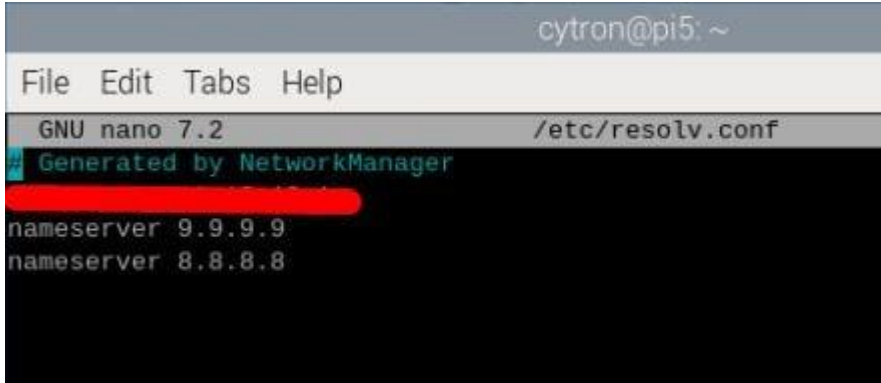
7.4 หลังจาก 4G module ทำงานในโหมด ECM แล้ว อันดับถัดมาให้ทำการตรวจสอบว่า ตอนนี้ตัว 4G module ได้มีการจ่าย IP address ออกมาให้กับตัวบอร์ด Raspberry Pi แล้วหรือยัง โดยใช้คำสั่งด้านล่าง

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                        |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|
| <b>ip a</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | โดยหากตัวบอร์ดมีการจ่าย IP address ออกมาแล้ว จะปรากฏ IP address ของ interface usb0 หรือ eth1 ด้านหลังคำว่า inet ดังภาพ |
| <pre>cytron@pi5:~\$ S ip a 1: lo: &lt;LOOPBACK,UP,LOWER_UP&gt; mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000     link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00     inet 127.0.0.1/8 scope host lo         valid_lft forever preferred_lft forever     inet6 ::1/128 scope host noprefixroute         valid_lft forever preferred_lft forever 2: eth0: &lt;NO-CARRIER,BROADCAST,MULTICAST,UP&gt; mtu 1500 qdisc pfifo_fast state DOWN group default qlen 1000     link/ether 2c:cf:67:63:02:ec brd ff:ff:ff:ff:ff:ff 3: wlan0: &lt;BROADCAST,MULTICAST&gt; mtu 1500 qdisc noop state DOWN group default qlen 1000     link/ether 98:b0:dc:f7:23:a6 brd ff:ff:ff:ff:ff:ff 4: eth1: &lt;BROADCAST,MULTICAST,UP,LOWER_UP&gt; mtu 1500 qdisc pfifo_fast state UP group default qlen 1000     link/ether 98:b0:dc:f7:23:a6 brd ff:ff:ff:ff:ff:ff     inet 192.168.225.28/24 brd 192.168.225.255 scope global dynamic noprefixroute eth1         valid_lft 42798sec preferred_lft 42798sec     inet6 fe80::3d19:bc56:bc7c:e32/64 scope link noprefixroute         valid_lft forever preferred_lft forever</pre> |                                                                                                                        |

แต่หากยังไม่ได้รับ IP address จาก 4G module สามารถแก้ไขได้โดยใช้คำสั่งด้านล่างนี้

|                      |                               |
|----------------------|-------------------------------|
| <b>sudo dhclient</b> | แล้วตรวจสอบอีกครั้งด้วยคำสั่ง |
| <b>ip a</b>          |                               |

7.5 โดยหลังจากตัวบอร์ด Raspberry Pi ได้รับ IP address จาก 4G module แล้ว สามารถทดสอบการเชื่อมต่ออินเทอร์เน็ตผ่าน 4G ได้ (อย่าลืมปิด WiFi หรือถอดสาย LAN ออกจากบอร์ดก่อนทดสอบ) โดยใช้คำสั่ง

|                                                                                     |                                                                                                                                     |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| <p><i>ping -c 4 8.8.8.8</i></p> <p>หรือ</p> <p><i>ping -c 4 google.com</i></p>      | <p>ในกรณีที่ ping ไปที่ Google ไม่ผ่าน แปลว่า DNS ตั้งค่าไม่ถูกต้อง สามารถแก้ไขได้โดยใช้คำสั่ง</p>                                  |
| <p><i>sudo nano /etc/resolv.conf</i></p>                                            | <p>แล้วใส่ DNS server ดังนี้</p> <p><i>nameserver 9.9.9.9</i></p> <p><i>nameserver 8.8.8.8</i></p> <p><i>nameserver 1.1.1.1</i></p> |
|  |                                                                                                                                     |

หลังจาก Save และ Exit แล้วทดลอง ping google.com ใหม่อีกครั้ง

**หมายเหตุ:** การเชื่อมต่ออินเทอร์เน็ตผ่าน 4G บริเวณ Taskbar ที่แสดงสถานะการเชื่อมต่ออินเทอร์เน็ตทางด้านขวามือจะไม่แสดงสัญลักษณ์ที่บ่งบอกถึงการเชื่อมต่ออินเทอร์เน็ต เนื่องจาก Raspberry Pi OS ไม่ได้ถูกออกแบบมาให้แสดงผลการเชื่อมต่ออินเทอร์เน็ตผ่านเครือข่าย Cellular แต่ทั้งนี้สามารถใช้งานอินเทอร์เน็ตผ่านสัญญาณ 4G ได้ตามปกติ